IEEE CLIMATECHAIN GLOBAL HACKATHON 2026 · CLIMATE DATA & ENVIRONMENTAL MONITORING

Afterburn

Farm-fire verification from the burn scar, and a pact that pays Punjab's farmers for not burning. Every verdict can be rechecked from public data.

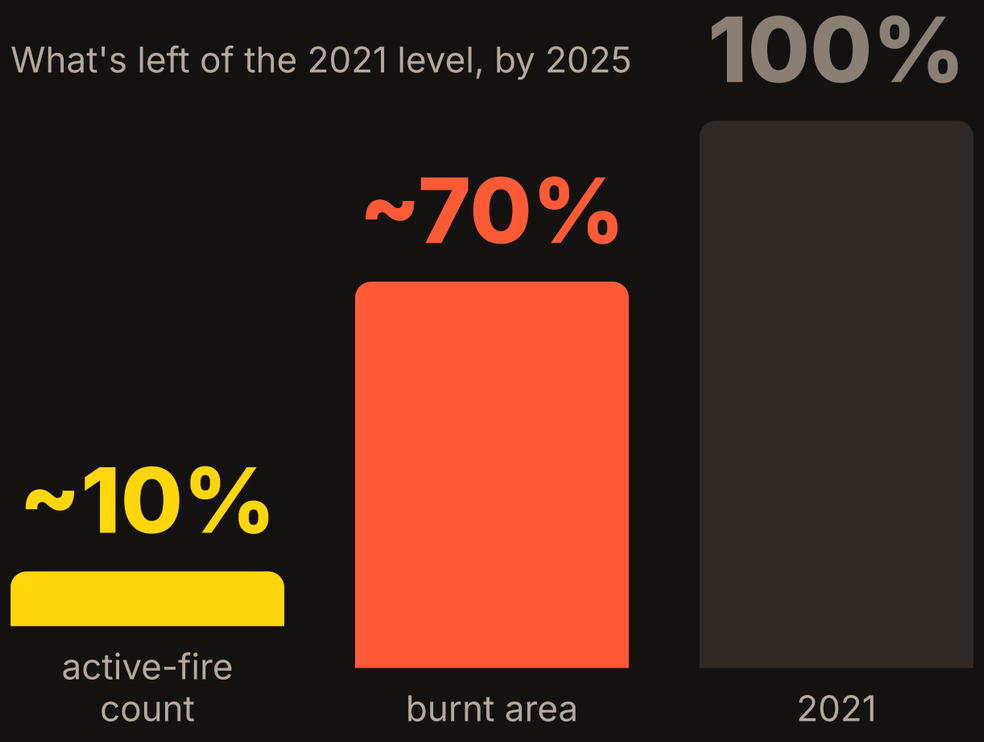
Jannet Ekka

github.com/JannetEkka/afterburn · jannetekka.github.io/afterburn

THE PROBLEM

Punjab's fire count fell 90%. Burnt area fell a third.

What's left of the 2021 level, by 2025



Paddy straw burnt every October and November is a large share of Delhi's winter smoke.

The official measure is the count of fires seen by NASA's MODIS and VIIRS, which pass at **10:30 am and 1:30 pm**.

Schemes that pay or fine farmers on that count now pay for nothing.

WHY

The fires moved to after the satellite passes.

>90%

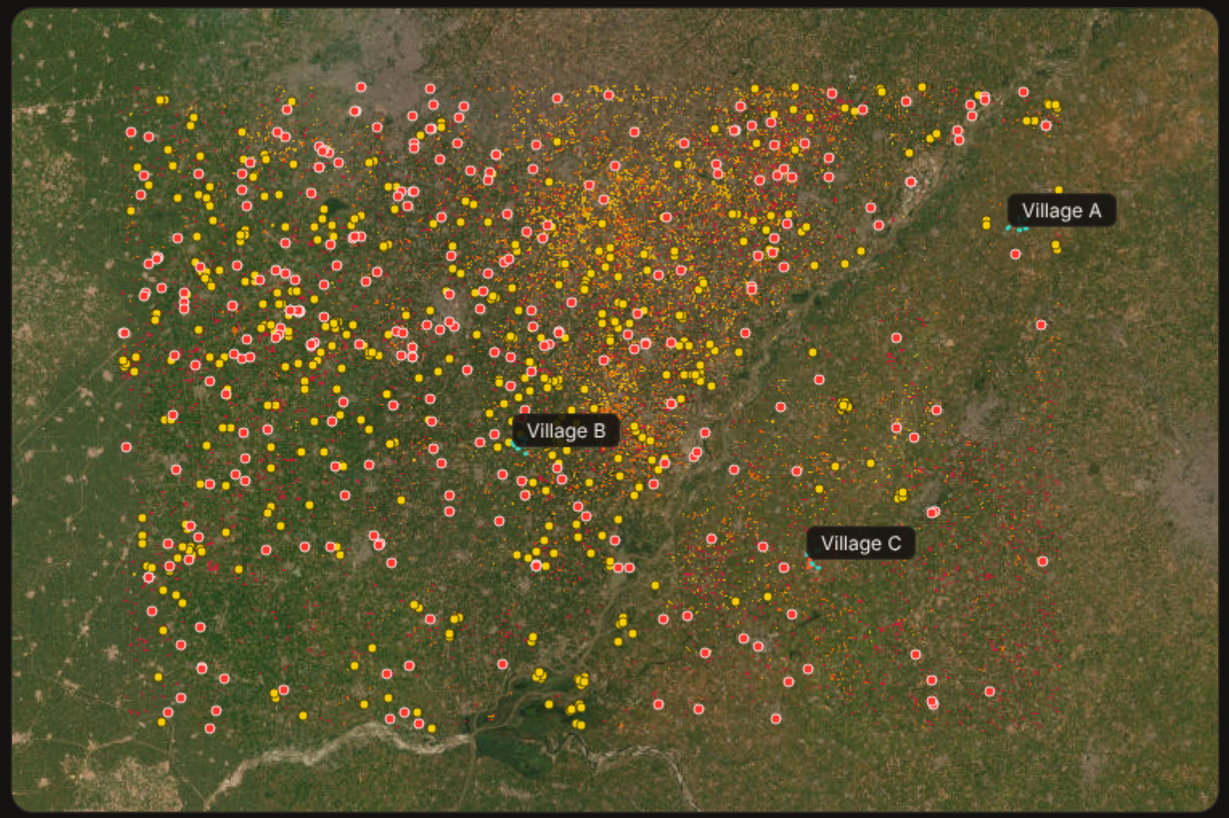
of large farm fires in Punjab in 2024 and 2025 started **after 3 pm**, seen by geostationary satellites that image every 15 minutes.

3%

did in 2021. NASA researchers found the same shift with GEO-KOMPSAT-2A. The fire is gone by the next morning's pass. The scar isn't.

iFOREST 2025 (Meteosat SEVIRI); NASA / H. Jethva, Dec 2025

THIS WEEK, IN OUR DATA



160

fields Sentinel-2 caught in flames at 11:20 IST on Oct 18. Only **8%** had a VIIRS detection within 1 km that day.

5,585

burn scars between the Oct 13 and Oct 18 passes. Only **2.9%** have a VIIRS detection within 375 m.

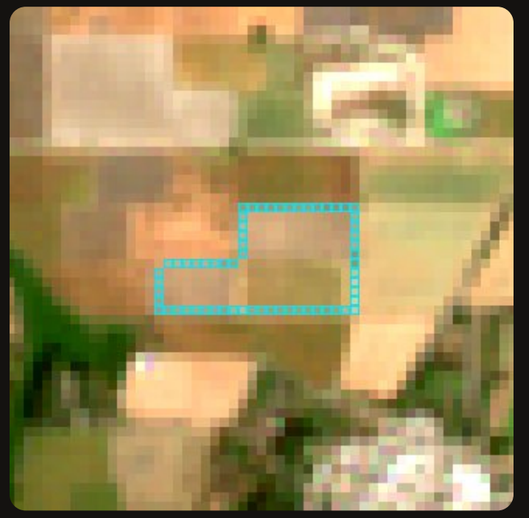
80 × 52 km of Amritsar, Tarn Taran and Kapurthala. **23,963 ha** of scars since Sep 23.

Copernicus Sentinel-2 L2A (Earth Search) · NASA FIRMS VIIRS

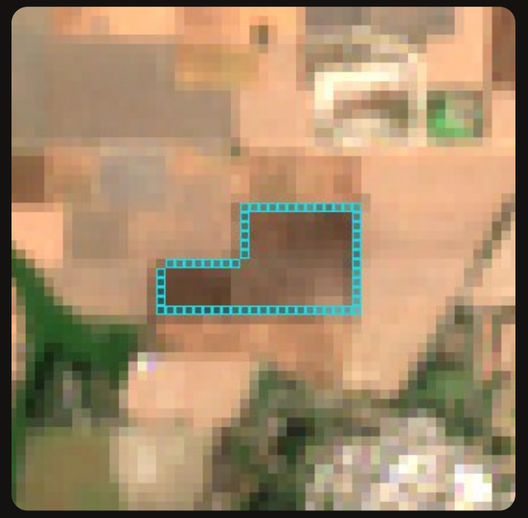
THE CHECK

Read the scar, not the flame.

No green crop	NDVI after < 0.35
Char signal	MIRBI rises ≥ 0.25 and reaches ≥ 1.6 (ISRO NRSC's burnt-area index)
Darker	SWIR-1 drops ≥ 0.02; visible ≤ 0.10; NIR ≤ 0.25
Not noise	3×3 median, ≥ 0.4 ha patch
Clear view	Sentinel-2 scene classes; hazy passes skipped
Burnt field	≥ 20% of it scarred over the season



Plot 19, Oct 13: stubble



Oct 18: 75% scarred

Each field is compared with its own last clear view, so a cloudy pass doesn't hide a burn. A dozen published thresholds, so anyone can recompute the result exactly.

DOES IT WORK?

Checked against a different satellite on a different schedule.

3.4×

Inside the 375 m footprint of each VIIRS fire detection, the rule calls **18.9%** of harvested land newly burnt. On random harvested land: **5.6%**.

2.5×

A positive-unlabeled gradient-boosting model trained on fields Sentinel-2 caught in flames, against **3.7×** for the rule in the same test. The rule makes the verdicts.

Honest limit: this shows the rule finds burns. It isn't a per-field accuracy figure. The best published Punjab model reports a Dice score of 0.64, so a pilot needs ground checks on a sample.

THE PACT

Built on what worked in a 171-village trial.

Paying part of the reward upfront raised compliance by **10 percentage points** (Jack, Jayachandran, Kala, Pande, *Money (Not) to Burn*, J-PAL).

Fund

A state scheme, CSR budget or straw buyer deposits. The contract won't promise money it doesn't hold.

Enrol

The cooperative signs a field up. 25% paid now. Each 50 m land cell can be enrolled once.

Verdict

Share of the field scarred + the evidence file's hash, posted after the season.

Challenge

Anyone who recomputes a different answer stakes a bond. The arbiter rules.

Settle

Clean: the other 75%. Burnt: nothing more, money freed for the funder.

Village bonus

Only if every enrolled field stayed clean. Neighbours talk each other out of it.

contracts/AfterburnPact.sol · 13 tests

LIVE ON MONAD TESTNET

24 real fields, a whole season, and a verdict anyone can recheck.

```
$ python pipeline/recheck.py docs/evidence/plot-19.json --
pact 0xC892…6b2D --plot 19
recomputed: 74.55% of the plot scarred, between 2026-10-13
and 2026-10-18
evidence hash, recomputed: 0x8cb26d12…bac7b26b
evidence hash, file:       0x8cb26d12…bac7b26b
evidence hash, on-chain:   0x8cb26d12…bac7b26b (74.55%)
MATCH: the verdict stands on public data.
```

- **Village A**: 8 fields clean, bonus paid
- **Village B**: 1 of 8 burnt, bonus lost
- **Village C**: 3 of 8 burnt, bonus lost
- A wrong challenge on a burnt field: rejected, bond to the pool
- Farmers get the result in Punjabi and English (Gemini)

WHO USES IT, AND HOW IT SCALES

Funders: Punjab and Haryana's residue-management schemes, CSR budgets, and bio-CNG and biomass plants that need straw collected, not burnt.

Cooperatives and NGOs enrol farmers and help them challenge a verdict.

Regulators (CAQM) get burnt area that doesn't depend on the hour of the fire.

Scale: Sentinel-2 covers all of Punjab and Haryana every 2–5 days, free. One 80 km area runs in minutes on a laptop.

Adoption: payments can stay on the existing direct-benefit-transfer rails, with the pact as the public ledger they settle against.

LIMITS AND NEXT STEPS

What it can't do yet

- A field burnt and ploughed before the next clear pass can slip through. The rule leans toward clean.
- No field-level accuracy figure without ground visits.
- Clouds and smoke haze cost passes (2 of 6 here).
- Punjabi notices not yet checked by a native speaker.

Next

- One cooperative, one funder, one season: ground-check a sample of verdicts.
- Add geostationary fire data as challenge evidence.
- Arbiter as a multisig: funder, farmers' union, independent scientist.